Screen Detection from Egocentric Image Streams Leveraging Multi-View Vision Language Model

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Abstract

Accurately monitoring the screen exposure of young children is important for research related to screen use, such as childhood obesity, physical activity, and social interaction. Most existing studies rely upon self-report or manual measures from bulky wearable sensors, thus lacking efficiency and accuracy in capturing quantitative screen exposure data. In this work, we developed a novel screen detection framework that utilizes egocentric images from a wearable sensor, named the screen time tracker (STT), and a vision language model (VLM). In particular, we devised a multi-view VLM that takes multiple views from egocentric image streams and interprets screen exposure dynamically. We validated our approach by using a dataset of children's free-living activities, demonstrating significant improvement over existing methods in conventional vision language models and object detection models. The combination of a vision language model and a lightweight hardware design provides a novel solution in screen detection for children. The proposed framework has great potential to benefit children's behavioral study. The code is available at https://github.com/YGanLab/MV-VLM.

Index Terms—

Vision Language Model; Wearable Sensor; Egocentric Image Streams

I. Introduction

Screen exposure has garnered increasing attention in the past decades due to the dramatic rise in digital technology. Extensive screen exposure has been associated with health and psychological problems [1] such as eye problems [2], language disorders [3], sleep disorders

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[4], obesity [5], and cognitive impairments [6]. In particular, following the suggestions from the World Health Organization (WHO), children between age between 2 and 4 should have less than 1 hour of screen exposure time per day [7]. However, this age group has been reported to spend an average of 2.5 hours in front of screens [8]. Therefore, it is important for parents to objectively monitor children's screen exposure and manage children's screen activity. In addition, there is also an unmet need for scientists to accurately measure screen time to better understand the association between health concerns and screen exposure [9]. Existing research methods for measuring screen exposure rely on users' self-reporting, experimental technologies (e.g., eye-tracking glasses), or built-in device apps. Whereas self-reporting is prone to bias, eye-tracking glasses can be invasive, and built-in apps can't measure exposure across devices or confirm users' identity. There is currently no non-invasive and automatic solution for accurate and robust measurement of screen exposure on cross-device screens.

A wearable camera with the capability to capture egocentric images is a promising non-invasive candidate to monitor screen exposure. Egocentric images are captured from a first-person perspective and can demonstrate the wearer's viewpoint. Egocentric images have been utilized for human activity analysis [10]–[12]. Although there is existing work that uses wearable cameras to track children's screen exposure [13], [14], those solutions are not ideal due to their bulky size and lack of automated analysis. Artificial intelligence (AI) enables computers to automatically identify objects with high accuracy, making it promising for screen exposure measurement. While AI can identify object types from egocentric images, no automated algorithm has been customized for children's screen exposure. Recently, vision language models have shown great potential for robustly identifying objects and events from images [15]–[17]. However, no such efforts have been made specifically for egocentric images for screen type identification. Convolutional neural network (CNN), as a major branch of AI, has advanced automated object detection methods in various applications by providing the backbone architectures for the development of the regional based CNN (R-CNN) [18] and You Only Look Once (YOLO) [19] detection systems. However, such methods work on individual frames and relies on local features extracted from CNN, lacking the capability to associate long-range dependency of features or high-level objects.

In this paper, we proposed a combination of an egocentric camera, namely the screen time tracker (STT), and a vision language model (VLM) to identify screen existence among multiple screen devices. The STT device and representative images are shown in Fig. 1 a). We developed a multi-view vision language model (MV-VLM) to process sequential frames from egocentric image data, as illustrated in Fig. 1 b). Specifically, we designed a multi-view scheme that captures temporal features across image sequences, enhancing the model's ability to detect screens even under occlusion. In parallel, we incorporate a pre-trained large language model (LLM) to generate human-readable explanations for each image group. These explanations reveal the spatial relationships among objects within the scene, particularly those relevant to the identification of a screen. We took advantage of a customized wearable sensor and egocentric images to identify children's screen exposure. In comparison with existing single view input vision language models, we developed a unique multi-view vision language model that process the egocentric image stream from wearable devices. Notably, a contrastive learning-based view selection module and a screen type identification module are specifically designed to address the challenges in existing methods and to develop a robust approach to detecting children's screen exposure. The proposed approach innovatively takes multi-view rather than single-view images as input compared to the existing screen identification work [20], [21]. The innovation contributes to the model's capability in extracting long-dependent textual features alongside spatial-temporal features from multi-view imaging, achieving the highest performance among all three types of screen scenarios.

Our major contributions are summarized as follows:

- We proposed a wearable sensor solution for identifying children's screen exposure. Our lightweight wearable sensor is child-friendly and captures egocentric images to monitor children's electronic screen activities. We collected data from children's free-living activities over two days, the first of its kind for egocentric cameras.
- We designed the first vision language model to identify children's screen exposure using wearable sensors. This model processes egocentric images to generate descriptive content, from which screen type identification is achieved through keyword extraction from the generated text.
- We devised an MV-VLM model, which is specifically for analyzing image stream from wearable cameras. To leverage temporal variations and capture more spatial features for accurate type identification, we developed a novel model that takes multi-view inputs and fuses features from different views for language generation. The selection of multi-view images is based on contrastive learning, which maximizes the information among images from different views.

II. Related Work

Screen Exposure and Wearable Cameras.

Understanding the link between health issues and screen exposure is crucial, necessitating precise tracking of screen time. Several studies have explored the extent of children's screen exposure using wearable cameras, which are non-invasive and less dependent on bias compared to self-reporting [22], [23]. Wearable cameras also indicate that a majority distribution of screen exposure goes to TV (42.4%) of children's daily screen exposure in [24]. Overall, very limited work [22] has explored the solution to automate the screen detection from wearable cameras.

Traditional Neural Networks in Egocentric Videos.

However, while many of these works rely on manual annotations, an automated solution for data analysis could largely reduce human workload. Deep learning models have been largely adapted for processing the videos from the egocentric wearable camera. Chen et al. [25] utilize the convolutional neural network and random decision forest for activity recognition; Song et al [26] utilize CNN-based and VLP-based models in extracting surrounding information for visually impaired persons; Bock et al [27] utilize ActionFormer for outdoor sports recognition; [28] utilizes an LSTM-based encoder-decoder framework to predict movement trajectory of a targeted person. The integration of multi-modal data, such as inertial data [27] and IMU readings [28] demonstrated an improved performance compared to using vision-based images alone.

Multi-View Based Neutral Networks.

A multi-view deep learning model leverages data from multiple perspectives to enhance learning in tasks such as classification and recognition by integrating diverse sources of information. Its potential has been explored in recent research [29], [30]. Although it has been demonstrated in existing studies that the incorporation of multi-view in CNN-based models outperforms models with single-view [31], [32], the multi-view network has not been fully explored in processing egocentric image streams. A critical problem in applying multi-view networks is how to select multi-view images from the egocentric image streams.

LLM in Egocentric Streams.

LLM is a rising field, and incorporating LLMs into image processing has the potential to significantly improve upon the studies using traditional models such as CNNs. By leveraging the contextual capabilities of LLMs, researchers can achieve more accurate and comprehensive egocentric image understanding. LifelongMemory was proposed in [33] as a novel framework that uses multiple pre-trained models to answer queries from egocentric video content. Research in [34] addresses Ego4D natural language queries challenge with image and video captioning models. Most of the studies focus on answering queries, thus lacking the capability to systematically analyze daily life for specific applications. Moreover, there is a limited vision language model to address the identification of electronic screen type, even from a single view.

III. Method

The goal of our study is to identify children’s screen interactions by leveraging both temporal and spatial information from automatically collected egocentric image streams. We design a multi-modal model composed of four key modules: view selection, vision modeling, language modeling, and screen identification. The view selection module constructs multi-view image groups, capturing temporal visual context as part of the preprocessed dataset. Simultaneously, a pre-trained language model extracts environmental (spatial) features from associated captions. The vision model then converts the image groups into visual feature embeddings, while the language model transforms the captions into textual embeddings. These visual and textual embeddings are separately aligned and fused within a pre-trained LLM, which integrates both modalities to generate screen-related descriptions. Finally, keywords are extracted from these descriptions to identify instances of screen exposure.

A. Wearable Sensor and Data Collection

Following our previous work [35], the dataset was collected from STT, the core component of which was a miniature 5-megapixel camera equipped with a 120-degree wide-angle, gaze-aligned lens. The camera was attached to clothing via a magnetic clip mount, as shown in Fig. 1 a) (left panel), with a battery life exceeding 48 hours. Worn on the chest, the badge-like device captured egocentric images every 10 seconds at a resolution of 2592 × 1944.

Young children between the ages of 3 and 5 were eligible to participate in data acquisition. The experimental protocol was approved by the local Institutional Review Board at the home institutions. We collected data from 30 participants. Each participant wore the device for two days. The whole dataset roughly includes children's participation in free-living activities. After the two-day activities, a survey on the comfort of the wearable device was conducted following a conventional protocol [20], [21]. The screen exposures related to TV, computers, and smartphones were captured. Representative images are shown in Fig. 1 a) (right panel). Image data were manually labeled by an undergraduate research assistant.

B. Framework

Our framework collected multiple images from different views to identify the existence of electronic screens. As shown in Fig.2, we developed a novel screen identification framework that consists of a view selection module, a vision module, a language module, and a screen identification module. The core concept involves selecting and processing images from multiple egocentric viewpoints to enhance the robustness of electronic screen identification.

1) Conceptual Rationale for Multi-View Vision Language Model: As shown in Fig. 2, due to the motion of children during a screen event, the egocentric image stream usually includes views at various angles to the electronic screen. In the screen exposure scenarios, multi-view image processing could be beneficial in several aspects. For instance, when screens are only partially captured, multi-view images may provide complementary data from different parts of the screen, enabling deep learning models to better understand and integrate the screen's various structural components. Additionally, in cases of low-quality images, such as those that are blurry, it is challenging for deep learning models to classify the images. Utilizing multi-view images allows the model to synthesize features from various perspectives, thereby enhancing feature representation and improving classification accuracy. To identify the existence of electronic screens, it is essential to consider the spatial relationships among static objects like TVs, computers, windows, walls, and shelves. These objects maintain consistent spatial relationships in the real world, which are reflected in 2D images based on perspective principles. To capitalize on these features and their spatial interrelations, we employed a vision transformer coupled with a language model. This approach leveraged the model's capability to explore long-range dependencies in text descriptions and image processing.

2) View Selection: The image stream captured from STT is in a time series, representing objects from different views and timestamps. The term 'view' used in this paper is similar to 'perspective'. It refers to images captured from the same camera at different time instants, but within the same scene. This view selection module converts temporal change in image frames to spatial change in perspective/view. Images taken from the same scene with small variations of position, height, and orientation are considered similar, though environmental conditions, including light, time of day, etc., may change. We sought a mapping rule that is robust enough to reveal spatial relationships and partial or occluded screen information. To capture coherent image features, we chose a contrastive learning-based embedding approach to encode the images and analyze their similarity. Assuming the egocentric image stream $I_1, I_2, I_3, \ldots, I_n$, for each image I_i, we used contrastive language-image pre-training (CLIP) [36] to convert images to embeddings $CLIP(I_i)$. CLIP enables pairwise image similarity comparisons without requiring prior knowledge, making it more adaptable than heuristic methods like K-Means, which depend on predefined clusters and often struggle with the variability inherent in real-world screen detection scenarios. For any two images I_m and I_n, we measured the cosine similarity between two embeddings via:

$$Sim(I_m, I_n) = \frac{CLIP(I_m) \cdot CLIP(I_n)}{\|CLIP(I_m)\| \|CLIP(I_n)\|} \tag{1}$$

Where $(\cdot)$ represents dot product and $\|\cdot\|$ is the magnitude. Then, to split the image stream into multi-view image groups, we built a graph $G(V, E)$ to represent the images with their similarity according to values of $Sim(I_m, I_n)$. In Graph G, each node v_i represents an image I_i. V corresponds to the set of nodes (i.e., images) and E corresponds to the set of weighted edges that connect each pair of nodes within a time window. For two nodes v_m and v_n, the edge $E_{m,n}$ between them is *valid* only if $Sim(I_m, I_n)$ is within a range between an upper bound τ_h and a lower bound τ_l. The upper bound ensured that static images in consecutive frames would not be selected. The lower bound ensured these two images could capture the same screen-related scene. We built an undirected graph and selected connected components of size k as multi-view image groups. To get the largest number of multi-view groups, we first sorted all connected components of size k according to the sum of edge degrees in each k-size component in ascending order. Then we greedily selected and split the components starting from the components with the least edge degrees from G, and updated G accordingly. The detailed procedures are illustrated in Algorithm 1.

Algorithm 1

View Selection

Input: Graph $G(V, E)$

Output: Set of multi-view image groups (S)

Parameters: Valid components (Sv);

$Sv, S \leftarrow []$

for $E_{i,j} \in E$ **do**

  if $E_{i,j}$ is *not valid* **then**

    Delete $E_{i,j}$ from E

  end if

end for

for Connected component $g \subseteq G$ **do**

Append g to Sv

end for

Sort Sv

for Connected component $g \in Sv$ **do**

if $g \subseteq G$ **then**

Append g to S

Delete g from G

end if

end for

3) Vision Language Model: The VLM model includes three components: a ViT model for vision embedding, a MiniLM model for text processing, and a Llama model for text generation. The three modules are connected by alignment layers.

Visual Embedding (ViT). We utilized the Swin Transformer [37] for image embedding. The transformer model used in this paper is a pre-trained model on a conclusive natural image dataset [38]. Images were divided into smaller non-overlapping patches and learnt by the transformer blocks. The transformer blocks operated in a hierarchical manner, and each block applied multi-head self-attention mechanisms based on shifted windows. These shifted windows enabled a set of self-attention models to dynamically learn and extract image features, which include long-range dependence. In our model, the features extracted from different views were embedded to a fixed length and sent to the fusion layer. In the Feature Fusion layer, features with the same length were normalized and stacked. Then, an alignment layer, which was built by a fully connected layer, fused them into a constant dimension. In particular, the features were averaged to create an input for the following vision-language generating task. This process fused multi-view images equally to enhance a complementary integration of features extracted from the Swin Transformer.

Text Embedding (MiniLM). We employed MiniLM [39] to extract textual features from the captions associated with multi-view images, leveraging its ability to map input texts of varying lengths into fixed-length embeddings and to generate high-level semantic representations. MiniLM is made of a set of distilled self-attention models in the transformer implementation. In this study, our analysis was constrained by a limited sample space, as all egocentric images were sourced from indoor home environments. Furthermore, the sample set mainly comprises screen-related instances. In addition, the textual information involved in this study was mainly that relevant to screen events. This made MiniLM particularly well-suited for our application, as its distilled architecture efficiently captured the necessary textual embeddings during the training phase. In the Feature Synthesis layer, text embeddings were normalized and concatenated to ensure the brevity of image captions. These concatenated embeddings were then processed through a linear alignment layer, ensuring dimensional consistency with the embeddings used in the Vision model. In particular, the text embeddings extracted from this phase serve as the annotation for the following text generation model.

Alignment Layer. The alignment layer is a trainable, fully connected layer that projects multi-modal embeddings into the input space of the LLM. The visual embeddings are fused first, then processed by a visual-to-textual alignment layer. The textual embeddings are synthesized and passed through a textual-to-textual alignment layer. The resulting representations are subsequently fed into the LLM for screen identification.

Text Generation (Llama). In this study, we used a large language model, $Llama2-7B$ [40], to efficiently produce fast scene descriptions regarding the contents in the multi-view images. We used one set of fully connected layers, which served as a soft prompt, to align the visual features with the large language model. Similarly, another set of alignment layers connected the text embeddings with the large language model. The training of the text generation model was an optimization process for alignment layers, where θ_v and θ_t corresponded to the weights in alignment for the vision model and the weights in alignment layers for text mode. The loss function for the report generation task was jointly optimized by the combination of loss terms from each task:

$$L_{report}(\theta_v, \theta_t; X_v, X_t, X_p, X_r) = -\sum_{l=1}^{M} logp_{\theta_v, \theta_t}(x_l; X_v, X_t, X_p, X_{r<l}) \quad (2)$$

where x_l is a variable related to the predicted token, M represents the length of the generated text, X_r represents the current prediction text, X_v represents the visual embedding, X_t represents the textual inputs, X_p represents prompts, and $X_{r<l}$ represents the token before the predicted token. This loss function considers textual input, visual input, and tokens generated, thus seeking an optimized parameter setting to generate reliable scene descriptions.

4) Screen Type Identification: The text description of the multi-view images was further processed to identify screen types. First, we extracted keywords from the generated description. The keywords included a whole set of screen-related objects, such as “TV”, “Television”, “Cellphone”, “Smartphone”, “Monitor”, “Computer”, “Laptop”, etc. Second, keywords were categorized into major types through a look-up table, as shown in Table I. Keywords that fell into the same categories were combined and identified as the same screen type. For example, “Smartphone” and “Cellphone” were considered as “Smartphone”. “Computer Monitor” and “Laptop” were both considered as “Computer”. Due to the similarity between hand interaction with a screen, we combined tablet and smartphone into the same category.

IV. Experiments and Results

A. Dataset

The dataset has 1,800 images, corresponding to 600 groups of multi-view images, including different screen types of TV, smartphone, and computer in free-living environments.

The free-living environment encompasses scenarios with varying lighting conditions. The detailed distribution of screen types is presented in Fig. 3, and it aligns with screen usage patterns reported in [24]. The dataset includes 375 images of televisions, 270 of smartphones, 255 of computers, and 900 non-screen images. Among the 30 subjects, all contributed TV and non-screen images, while 15 subjects contributed smartphone images and 12 contributed computer images.

To build a multi-view image dataset with captions describing the screen of each group, captions were generated with guidance from Bootstrapping Language-Image Pre-training (Blip-2) [41] and indoor scene understanding [42]. The generated captions provide environmental context. However, the captions are not intended for detecting screen types. Rather, these captions capture spatial relationships between screens and surrounding objects, helping to address challenges posed by imperfect imaging conditions such as occlusion, blurriness, and poor lighting. To evaluate the model's performance across different screen types, we randomly shuffled the dataset and divided it into four folds, conducting cross-validation for robust validation.

B. Training Strategy for Implementation

The training process required minimal effort to customize the VLM for screen detection. In particular, we chose pretrained models using Swin Transformer for ViT [37] and MiniLM for the language model [39]. We fine-tuned ViT using our multi-view image datasets. In text generation, the Llama model ($Llama2-7B$ [40]) remained frozen as well. The training remained within the parameters of the alignment layers. The experiments were conducted in parallel on two NVIDIA H100 GPUs. We set the batch size as 6 and the learning rate as 0.0001. In multi-view selection, the size k of the multi-view group was set as 3. We also empirically set τ_h as 70% and τ_l as 40%.

C. Multi-View Images Selection

Our model selected images according to CLIP embedding. We picked a subset of egocentric images to input to VLM. A representative example of each screen type is shown in Fig. 4. We noted that the selected images complemented each other in terms of field of view and spatial features, demonstrating that CLIP was an effective feature extractor for evaluating similarity and selecting views. Moreover, we analyzed the CLIP-generated features using t-distributed Stochastic Neighbor Embedding (t-SNE) [43], to visualize the distribution of typical multi-view images. It is observed that not only are the features from the same multi-view group complementary to each other, but also the features among screen types are distinctive from each other. Such observation lays a foundation of text generation and screen type identification.

D. Text Generation and Screen Type Identification

We evaluated the generated text description from Llama with the ground truth using BiLingual Evaluation Understudy (BLEU) [44]. BLEU scores evaluate the similarity of the generated description by Llama and the reference caption. A higher BLEU indicates higher similarity to the ground truth. BLEU N, where N corresponds to contiguous sequences of

N word, indicates an evaluation of quality in a context. Moreover, our MV-VLM model was able to generate scene descriptions with logic and smoothness.

Figure 5 shows representative scene descriptions in each screen type category. Key words from each scene description are highlighted in red and categorized into the three screen types (i.e., TV, smartphone, and computer). ChatGPT4 [45] is utilized to smooth the description for scene understanding. Our vision language model effectively identified various screen types, their spatial relationships, and action events with objects. As shown in the first row, our model successfully identified a computer even when its screen was partially occluded. In the second row, our method correctly detected a typical TV in a living room, despite the screen being distorted through a plastic bottle. In the final row, where images were captured during motion, our proposed MV-VLM effectively distinguished a smartphone from the environment, even under blurry conditions.

E. Comparison with Existing Methods

To validate the superiority of MV-VLM over existing methods, we compared the performance of identifying screen types with multi-view Swin Transformer [46], YOLOv11 [47], LLaVA [48], and a modification of CLIP for the classification task. The results are reported in Fig. 6. The multi-view Swin Transformer (MV-Swin-T) is a Swin-Transformer-based algorithm that takes multiple images as input and classifies the input images. YOLOv11 is an object detection approach that uses a CNN as a backbone to output the bounding box and screen type. It is currently the state of the art that provides high accuracy in object detection. We concatenated the multi-view images for the YOLOv11 as inputs. LLaVA is a multi-modal model that takes image and text inputs for caption generation. CLIP for Classification (CLIP-fC) is a modification of CLIP by adding a fully-connected layer on top of CLIP features, enabling it for classification tasks. As shown in Fig. 6, our MV-VLM outperformed the existing state-of-the-art (SOTA) algorithms. Moreover, as shown in Fig. 5, MV-VLM identified spatial relationships among objects (e.g., laptop upon a wooden desk) and action events (e.g., holding a smartphone).

LLaVA takes image and text as input and generates captions, followed by the same keyword extraction and identification process. The performance of LLaVA is slightly lower by approximately 20%, compared to our proposed method. This highlights the importance of retraining a VLM with an alignment layer specifically tailored for screen type identification using environmental context. The superior performance of our model also shows the value of multi-view processing in VLMs. Unlike our approach, LLaVA processes only single-view images and completely ignores temporal information. Furthermore, as shown in the text generation results in Fig. 6, our model achieves higher BLEU scores than LLaVA, reinforcing the importance of fine-tuning VLMs for the specific task of screen identification. BLEU scores are reported only for VLM-based models, such as LLaVA and our proposed MV-VLM.

The degraded performance of MV-Swin-T and YOLOv11 may be attributed to their limited access to spatial information derived from language model components. In free-living home settings, children's environments often involve complex conditions such as occlusion and varying lighting. Under these circumstances, screens are difficult to accurately identify using vision-only models. Directly using visual models will fail to detect the screen. Space relationships (for example, TV on the wall or stand) are comparatively fixed in objects detected within home settings, making our approach a good solution to use spatial relationships to detect screens. In contrast, CLIP-fC leverages a pretrained language model alongside information from multi-view images, using majority voting to determine classification outcomes, and thus gains a better performance than MV-Swin-T and YOLO. Overall, vision language models such as MV-LVM, LLaVA, and CLIP-fC consistently outperform non-VLM approaches, highlighting the importance of incorporating VLMs for this application.

F. Ablation Study

We conducted an ablation study to investigate the contributions of components in model architecture and view selection strategy. The results are reported in Table II. To validate our model design, we conducted ablation studies by systematically removing four key components: the large language model, multi-view inputs, MiniLM, and the Swin Transformer. To assess the impact of removing the LLM, we replaced it with a fully connected layer to classify screen types based on visual and textual embeddings. In the absence of multi-view inputs, the model relied on single-view images and their corresponding environment descriptions for text generation and screen classification. When removing the MiniLM module, we directly wrapped the environment descriptions and visual embeddings into a prompt and passed it to the LLM. To evaluate the role of the Swin Transformer, we substituted it with a ResNet [49] backbone for visual feature extraction. The results of these ablations are presented in the Table II. As shown, our complete model consistently outperforms the ablated variants (TV: 0.9013±0.0121, Smartphone: 0.8753±0.0124, Computer: 0.9240±0.0292), indicating the importance of incorporating the LLM, multi-view image inputs, MiniLM, and Swin Transformer in the framework.

We further conducted an ablation study to evaluate different view selection strategies. In the absence of CLIP for view selection, we explored the following alternatives: (1) using temporally consecutive frames, (2) replacing CLIP with ResNet, and (3) replacing CLIP with K-Means clustering. The results are presented in Table II. As shown, our multi-view selection approach using CLIP embeddings achieves the best performance, highlighting its effectiveness.

G. Sensitivity Analysis on τ_h and τ_l

We conducted a sensitivity analysis to evaluate the impact of different values of τ_h and τ_l in the proposed view selection module. These thresholds—τ_h (upper bound) and τ_l (lower bound) play a critical role in determining the selection of multi-view frames. As shown in Fig. 7, setting τ_h to 70% and τ_l to 40% yields the highest F1 score and the second-highest average accuracy across different screen types. Notably, our view selection algorithm demonstrates strong robustness to variations in τ_h and τ_l.

H. Screen vs Non-Screen Classification

In behavioral studies, it is more important to determine the existence of electronic screens rather than identifying the screen types. To this end, we conducted a binary classification test to evaluate the performance of the MV-VLM in detecting the existence of screens or not. We had 600 groups of multi-view images in the test set. The results, including a confusion matrix, are presented in Table III. The model achieved an overall accuracy of 95.5% and a sensitivity of 93.67%, indicating a stronger capability in detecting the screen existence with a small fraction of false negative detection.

I. Assessment of Comfort on Wearable Device

We conducted an assessment of comfort on the wearable device. As shown in Table IV, the device receives high evaluation scores in emotion, harm, movement, and anxiety. The results suggest that the device is comfortable and acceptable for regular use by children. The high scores demonstrate the effectiveness of our tailored design for children, which includes a lightweight, magnetic clip mount for secure attachment, a smaller and lighter badge-like design, and customized shapes to appeal to children.

V. Discussion

In this study, we propose MV-VLM, the first known vision language model that is deployed for screen type identification. The dual extraction performance distinctly surpasses that of other existing models. The efficacy of MV-VLM in processing and integrating these diverse data types indicates potential utility in applications where a comprehensive understanding from multiple perspectives is crucial. Moreover, MV-VLM only requires a minimal amount of training samples, as the training is on the alignment layers and the other models are fine-tuned on pre-trained models (Swin Transformer and MiniLM). Such a feature enhances the feasibility of deploying our model in resource-constrained environments. The proposed algorithm has an average inference time of 0.3 seconds, opening the opportunity for real-time inference in the STT platform. This high efficiency of image analysis also demonstrates strong deployment feasibility. Besides the data analysis, we also improve the data collection process. Unlike previous studies that collected data from adults in controlled environments, this study focuses on data collected from children in free-living environments, making the dataset more representative of real-world usage and future behavioral studies. To protect children’s privacy, guardians and children may delete images that they feel uncomfortable with at the end of data acquisition, following conventional protocols in [20], [21].

This study has a few limitations. First, we only examine the case where a single screen type is involved. We will expand the framework to multi-task classification when multiple screens are involved in sequential images. Second, our data collection has been restricted to indoor environments, which limits the exposure of our model to varied environmental contexts. In future research, we plan to expand our data acquisition efforts to outdoor settings, thereby enriching the model’s training dataset with a broader range of environmental dynamics. This expansion is anticipated to boost the model’s generalization capabilities and enhance its performance across more diverse real-world scenarios.

To ensure a fair evaluation of the algorithm development, we exclude samples with severe motion blurring and images accidentally covered by clothes or other objects. We consider those images as outliers that do not correspond to features captured by wearable sensors. In the integration of a hardware system, we plan to use accelerometers to inform the exclusion of low-quality, blurry images and use ambient light sensors to instruct the system that there is occlusion and the multi-view images are not needed. We also plan to consider the proximity of the screen in our framework by leveraging the depth information [50]. Future study will seek to integrate behavioral research to the measurement obtained from MV-VLM. By correlating our model's outputs with behavioral data, we aim to deepen our understanding of the interactions between environmental contexts and individual behaviors. This approach is promising not only to validate MV-VLM but also to extend to more complex, behaviorally-driven studies. Our model shows high efficiency in image inference, whereas its overall parameters can be further reduced without much decline in performance by applying parameter-efficient techniques such as model distillation for better deployment feasibility. Lastly, we will consider applying the cross-attention mechanism to enhance the learning of spatial relationships between screens and other objects from both visual and textual features, further improving the performance of the proposed framework. The output of our algorithm is interpretable in human language, enabling downstream tasks related to the analysis of children's screen activity. For example, the algorithm can generate textual descriptions of screen-exposure events, which helps preserve the privacy of the children involved in the study.

VI. Conclusion

We proposed a lightweight and comfortable wearable sensor solution using a screen time tracker to monitor children's screen exposure. We devised a multi-view vision language model to identify the existence of screens from egocentric image streams, where data were collected from children's free living activities. We explored a multi-view image input scheme, which captures spatial-temporal features across image sequences, enhancing the features of screen identification. This multi-view model is integrated with a vision language model, generating image descriptions that are related to screen existence. Our experiments indicate superiority in comparison with conventional vision language models and object detection models. Future work will include cumulative measurements of screen time exposure and associate the screen measurements with other factors in behavioral studies.

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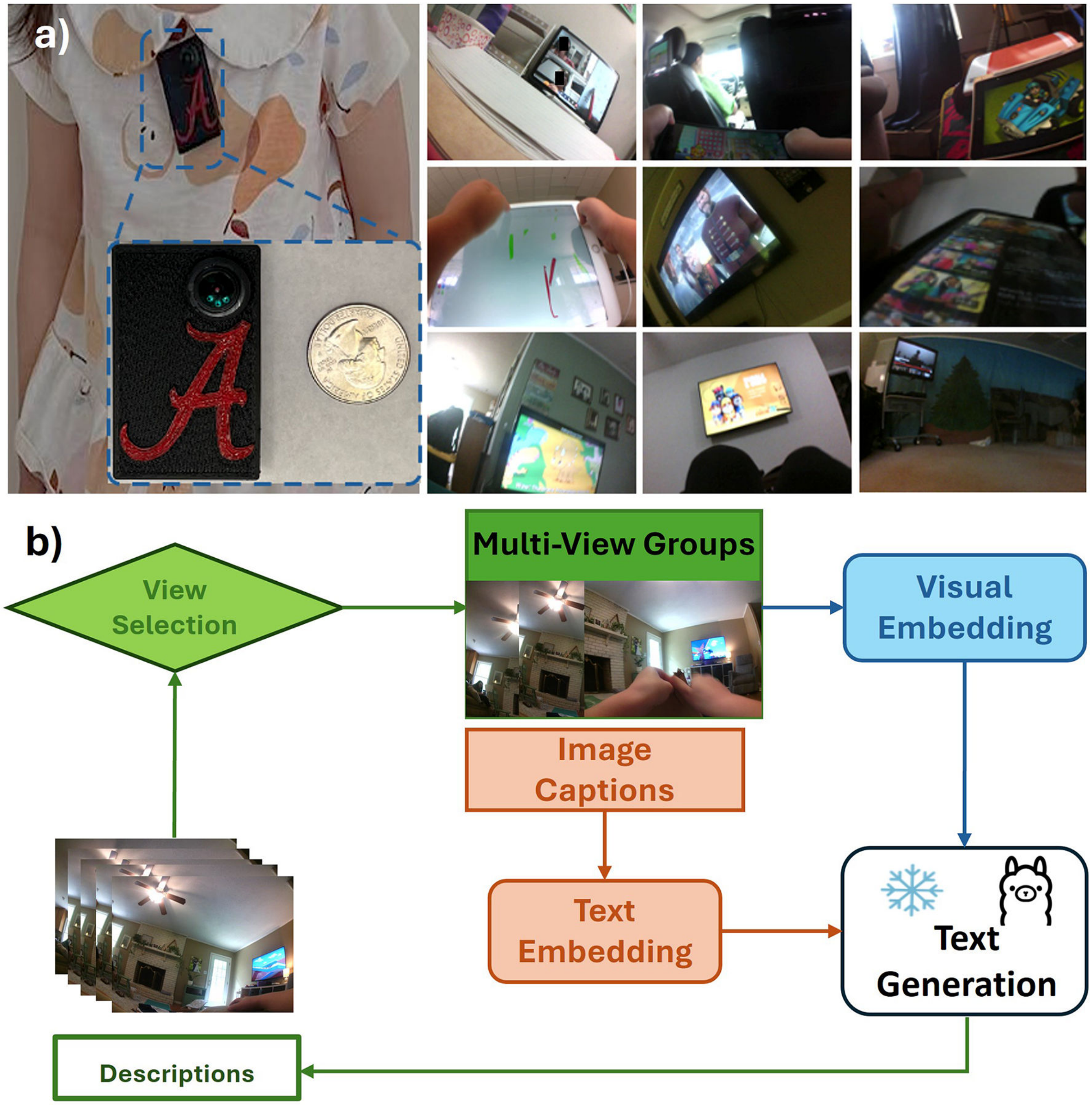

Fig. 1.

Overview of the study. a) The STT device (left panel); compilation of images showcasing various environments involving screen exposure. The montage is created using free-living data collected from STT (right panel). The STT device is lightweight and can be firmly attached to clothes. b) The architecture diagram of the proposed MV-VLM. Our study deploys both temporal and spatial features to enhance the screen detection capabilities.

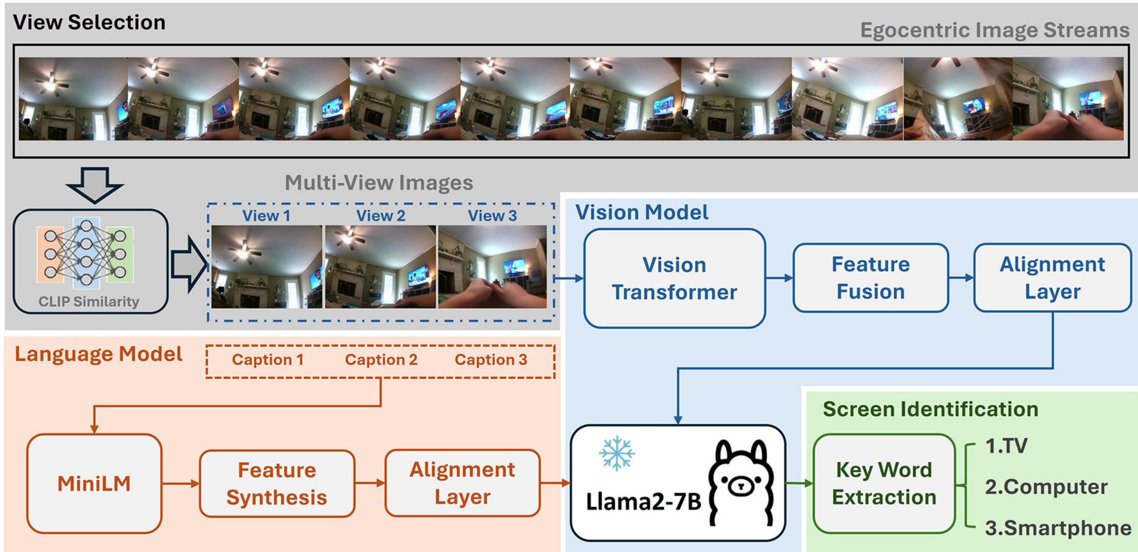

Fig. 2.

Proposed pipeline to process egocentric image streams. There are four major components: view selection, vision model, language model, and screen identification. The view selection module uses Contrastive Language-Image Pre-Training (CLIP) to extract embeddings and select multi-view images based on similarity. The vision language model learns from the vision transformer and MiniLM to generate textual descriptions for multi-view images.

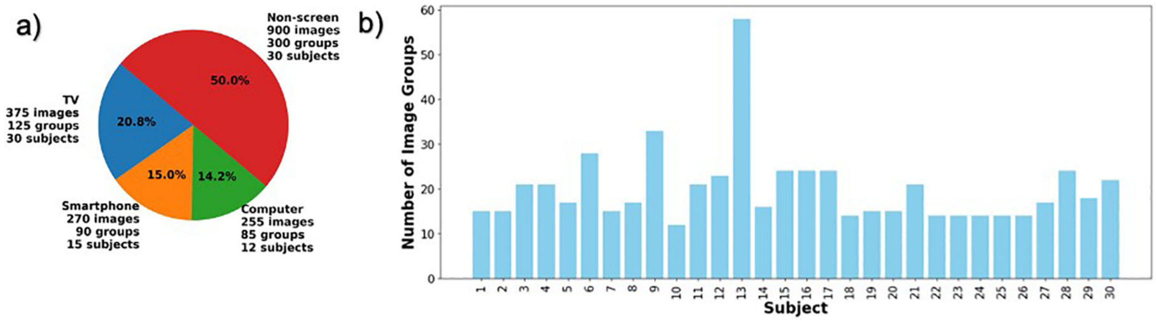

Fig. 3.

Overview of the dataset used in this study. a) Distribution of different screen types captured in the egocentric image dataset. b) Distribution of the number of image groups collected per subject.

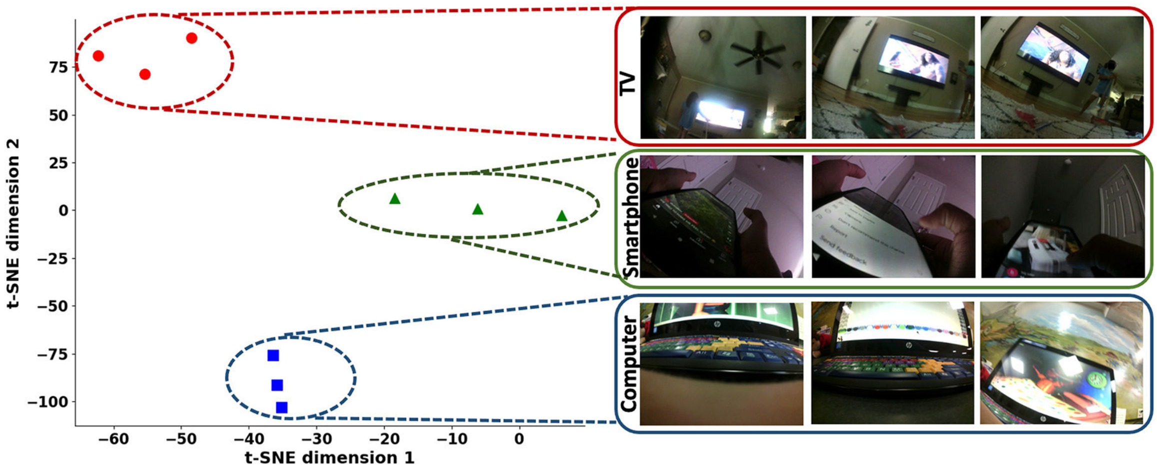

Fig. 4.
Representative multi-view images selected using CLIP embedding. Our selection maximizes variation among consecutive frames while capturing complementary features of screen objects from different views. The features are visualized using t-SNE.

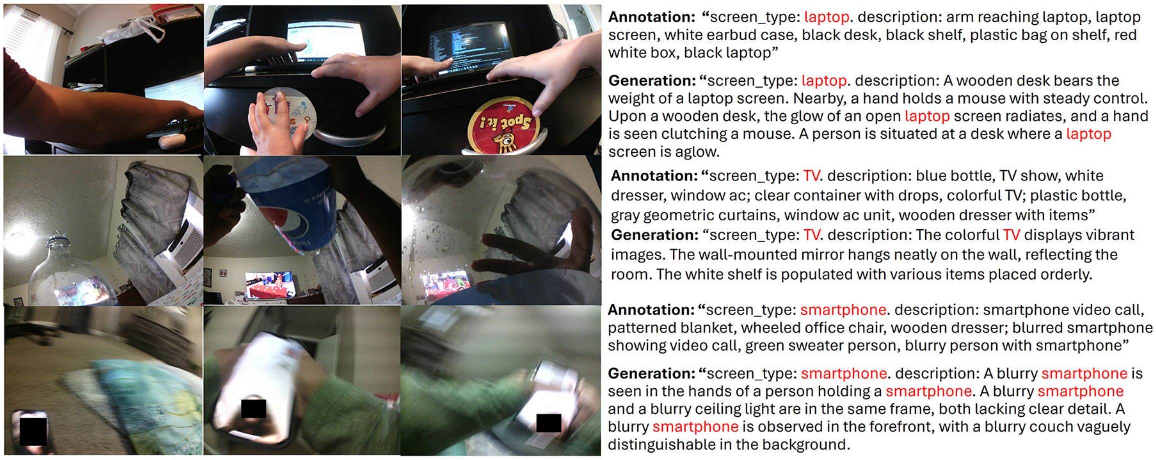

Fig. 5.

Examples of generated text and screen identification. The left panel shows typical multi-view images. The right panel shows the generated descriptions from the proposed model and the annotations. Key words could be efficiently processed to categorize images into a specific screen types.

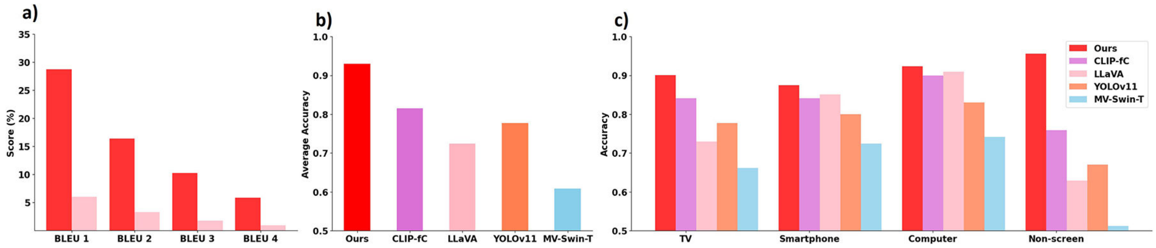

Fig. 6.

a) Text generation results from the proposed method (MV-VLM) and LLaVA to compare how closely the generated scene descriptions match the ground-truth annotations. b) Overall accuracy comparison among MV-VLM, CLIP-fC, LLaVA, YOLOv11, and MV-Swin-T. c) Comparison among MV-VLM, CLIP-fC, LLaVA, YOLOv11, and MV-Swin-T in the identification of TV, smartphone, and computer. Our method consistently shows a better performance than the other baseline methods.

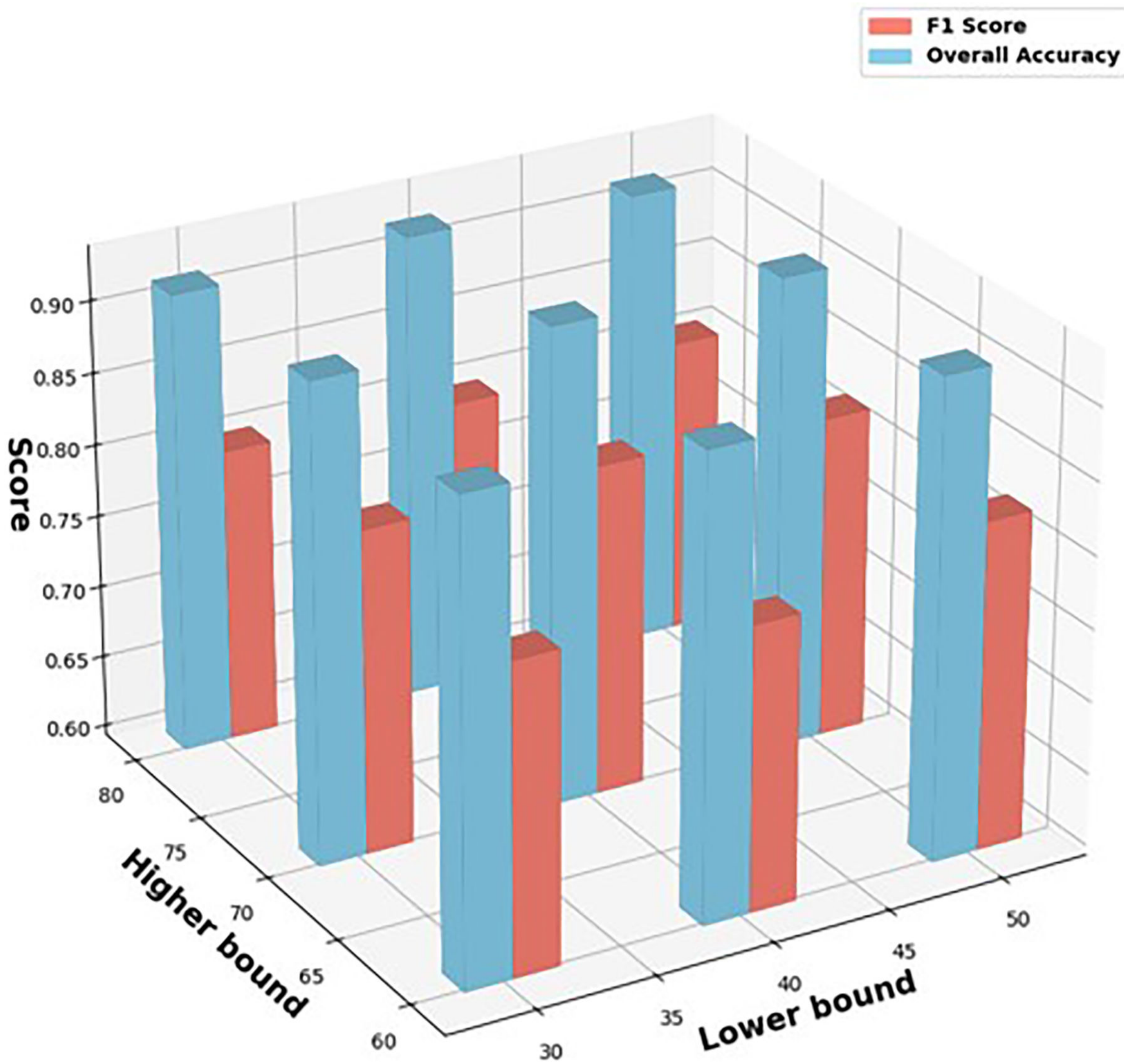

Fig. 7.
Sensitivity analysis of the upper threshold τ_h and lower threshold τ_l in the view selection module. The results highlight that our view selection algorithm maintains high performance across a broad range of $\tau_h - \tau_l$ combinations, demonstrating strong robustness.

TABLE I

A list of mappings between keywords and screen types

Screen Types	Key Words
TV	TV, Television
Smartphone	Smartphone, Phone, Tablet, Cellphone, iPad
Computer	Computer, Laptop, Computer Monitor

TABLE II

Performance comparison of model architecture ablation and view selection strategy ablation variants across different screen types. The best performance is highlighted in red and the second-best performance is highlighted in blue

Ablation Variant	TV	Smartphone	Computer	Non-Screen	Acc.
Ours	**0.9013**	0.8753	0.9240	**0.9561**	**0.9300±0.00929**
Model Architecture Ablations					
w/o LLM *(use fully-connected layer)*	0.8944	0.8660	0.8948	0.8417	0.8650±0.0398
w/o multi-view (use single image)	0.8651	**0.8813**	0.9198	0.8972	0.8917±0.00861
w/o MiniLM (use direct input text)	0.8614	0.8521	**0.9301**	0.9218	0.8967±0.0175
w/o Swin-Transformer (use ResNet)	0.7039	0.8063	0.8615	0.7077	0.7433±0.0512
View Selection Strategy Ablations					
Remove CLIP	0.8647	0.8515	0.8746	0.9525	0.9083±0.0238
Replace CLIP with ResNet	0.8661	0.8435	0.8777	0.9447	0.9033±0.0365
Replace CLIP with K-Means	0.8857	0.8470	0.9027	0.9299	0.9050±0.00812

TABLE III

Confusion matrix from binary classification. We combine the classes from any type of screen as “screen” and the rest as “non-screen”.

		Predicted Class	
		Positive (screen)	**Negative (non)**
Actual	Positive (screen)	281	19
Classes	Negative (non)	8	292

TABLE IV

Assessment of Comfort (All values are out of 10)

Metrics	Mean
Emotion (Is it acceptable for children to wear the sensor?)	8.23
Harm (The sensor does not cause pain or tickling.)	8.36
Movement (The sensor does not restrict children’s moving.)	9.23
Anxiety (Children feel secure wearing the device.)	8.2